

Assignment - 01.

CSA5113 -

Crypto Graphy.

Karthieeka Repalle 98/100

192372289

CSE - AI

1.

Encrypt the message "DEPEND THE WALL" using Caesar cipher with key=3. Decrypt the message "KHQR ZRUG" using Caesar cipher with key=3. A Caesar cipher is used to encrypt the word "SECRET" to "VHFUHW". Determine the key used, and decrypt "WYLVLDWH VW".

The Caesar is one of the simplest substitution cipher. Each letter in the plaintext is shifted by a fixed number of positions in the alphabet in this question.

a) Encrypt the message "DEPEND THE WALL" key=3

Plaintext : DEPEND THE WALL

Plain text :- D E F E N D T H E W A L L

Cipher text :- G H I H Q Q W K H Z D O O

⇒ GHIHQQ WKH ZDOO

b) Decrypt the message "KHQR ZRUG"

This decryption process shifts each letter 3 positions backward

Cipher text : KHQR ZRUG.

Plain text :- HELLO WORLD.

c) Determine the key used.

SECRET → VHFUHW.

S → V = +3

∴ K=3

Decrypt : WKLVLVD WHVW

Plain text : THIS IS A TEST.

2. Monoalphabetic Substitution Cipher :-

A monoalphabetic substitution cipher unplace each alphabet with another fixed alphabet throughout the message unlike the caesar cipher, every letter has its own substitution.

a) Encrypt "FIRE"

$A \rightarrow Q, B \rightarrow W, C \rightarrow E \dots Z \rightarrow M$

Plain : F I R E

Cipher : Y O R T

Cipher text : YORT

b) Frequency Analysis :-

Frequency analysis is technique used to buake Substitution cipher by comparing the frequency of letter is the ciphertext with common english letter

Cipher text:

U J H H D Q E D N R F Z H Y O L T K

Possible Plain text :

The Quick BROWN Fox.

3.

Vigenere Cipher:-

The vigenere cipher is a polyalphabetic substitution cipher that uses a keyword to encrypt the plaintext. Each letter is shifted according to the corresponding letter in keyword.

a) Encrypt "ATTACK AT PALACE" using key "LEMON"

Keyword : LEMON LEMON LEMON

Cipher text: LXFOPVEFRNHRM

b) Decrypt "LXFOPVEFRNHR"

Keyword : LEMON LEMON L E

Plaintext : ATTACK AT DAWN

c) Decrypt "QNXEPKMAEGTU" using key.

Plaintext :- GDTA LAG WA WPQ

4.

Hill Cipher:-

The Hill cipher is a matrix-based substitution cipher. It converts letters into numbers and uses matrix multiplication with modulo 26 arithmetic to encrypt the plain text.

Key matrix : $\begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$

a) Encrypt "HI"

$$H = 7$$

$$I = 8$$

After matrix multiplication and modulo 26

cipher text: TC.

b) Verification :-

For Message PO

$$P = 15$$

$$O = 14$$

After multiplication,

Result = JW

Hence, .

PO does not encrypt the ac using
the given matrix.

5. Transposition Cipher

A Transposition cipher rearranges the positions of letters instead of replacing them. The order of letters changes according to the given keyword

a) Encrypt "MEET ME AFTER LUNCH" using
Keyword "ZEBRHS".

Encryption: MEET ME AFTER LUNCH

Keyword: ZEBRHS

Cipher text: - E A N M T E F C U H E T R E I M

b) Decrypt:-

Cipher text:

E A N M T E F C U H E T R E L

Keyword: BOARD

Plain text: MEET ME AFTER.

6.

One - Time Pad (OTP):

The OTP is one of the most secure encryption method, it uses a random key that is the same length as the plaintext and if it is used once when the conditions are satisfied.

a) Encrypt "HELLO" using key "XMCKL"

* Convert letter into numbers.

Letter: H E L L O

Value: 7 4 11 11 14

* Key :-

letter:- X M C K L

Value: 23 12 2 10 11

* Add values (modulo 26)

Plain: 7 4 11 11 14

Key: 23 12 2 10 11

total: 30 16 13 21 25

mod 26: 4 16 13 21 25

* Convert back to letters :-

4 = E

16 = Q

13 = N

21 = V

25 = Z

cipher text :- EQNVZ

b) Explain why OTP is unbreakable :-

The One-time Pad is considered theoretically unbreakable under ideal conditions because

* the key completely random.

* Key length is equal to the message

length.

- * Every possible plaintext is equally likely
- * Without the correct key, it is impossible to determine the original message

7.

Columnar Transposition cipher:-

Columnar Transposition cipher encrypt a message by writing the plaintext in row under a keyword and reading the columns in alphabetical order and keyword.

a) Encrypt "WE ARE DISCOVERED" using keyword "ZEBRA".

Plaintext : WEAREDISCOVERED

keyword: ZEBRA.

Arrange the message.

Z	E	B	R	A
W	E	A	R	E
D	I	S	C	D
V	E	R	E	D

keyword order:

Z	E	B	R	A
5	3	2	4	1

Read columns in the order 12345

cipher text: EODASREIERCEWDV.

b) Decrypt the ciphertext

ciphertext :- EVLNEACD TKESE A Q R O F O J D E
E C U X W N

Keyword :- TRIAL

Arrange the ciphertext into columns using the key-
words order, restores the original column positions,
and read row-wise

Plain text:

WE ARE DISCOVERED FLEE AT ONCE

c) Encrypt "INFORMATION SECURITY" using keywords
"HACK"

Plain text : INFORMATION SECURITY

Keyword : HACK — length = 19

Final plaintext :-
INFORMATION
SECURITYX

column : 4

Rows = 5

padding = X

Arrange:-

H	A	C	K
I	N	F	O
R	M	A	T
I	O	N	S
E	C	U	R
I	T	Y	X

Cipher text :-

NMOCTHNUYRIELOTSRX

Frequency Analysis

Frequency Analysis is used to break a monoalphabetic substitution cipher by comparing letter frequency with common english letters.

Ciphertext: UZQSOVDHXMOPVGPVGPOZPEVS
GZWSZOPFPESXUZUETUCG

1. Frequency of letters:-

letter	frequency
Z	8
O	6
P	5
U	5
S	5

2. Probable mapping:-

cipher	plaintext
Z	E
O	T
P	A
U	O
S	I

3. Partial Decryption:- After applying the above mapping only part of the message can be identified

eg:- OETIT.....

4. Best Guess:-

It was discovered that message is secret.

9. Known - Plaintext Attack:-

A known-plaintext Attack is a cryptanalysis method where part of the original plaintext is already known. This information is used to find substitution mapping and decrypt remaining message.

Given, ciphertext:

BJTW QTW YMTW NXF XNRUQTWX

Known plaintext:- This is THE.

1. Find mapping

	Plain
B	T
J	H
W	I
Q	S
Y	T
M	H
N	A
X	R

2. Decrypt the ciphertext.

Using the above mapping, message becomes plain text

THIS IS THE A SIMPLE TEXT

3. Validation

FINAL Answer:

THIS IS THE A SIMPLE TEXT.

Vigenere cipher cryptanalysis.

The Vigenere cipher uses a repeating keyword to encrypt a message. If the key is unknown, it can be recovered using cryptanalysis techniques such as frequency analysis & pattern matching.

Given ciphertext

ZICVTWQNGR ZGVTWAV ZHH CQVGLMJ.

Step 1 :- Recover the key.

Using cryptanalysis probable key is

KEY

EEY

Step 2 :- Decrypt the ciphertext

cipher :- ZICVTWQNGR ZGVTWAV ZHH CQVGLMJ

key :- KEY KEY ~~KEY~~ KEY KEY KEY

Final Answer :-

Recovered key : KEY

Plain text : WE ARE DISCOVERED SAVE

YOURSELF.